Reproducing the Federal Reserve's SCF Tables with the scf Package

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ABSTRACT

This analysis attempts to validate the R package `scf` by reproducing official statistics. Below, I reproduce Tables 1 to 4 of the Federal Reserve's historical tables, derived from the public-use Survey of Consumer Finances, for every wave from 1989 to 2025. The R Markdown file that produces this report can be reviewed and downloaded at https://github.com/jncohen/scf/blob/main/validation/scf2025-validation.Rmd.

1 Purpose

The R package `scf` (Cohen 2026d) facilitates the correct analysis of the public-use microdata release of the U.S. Federal Reserve Board's (FRB) *Survey of Consumer Finances* (SCF) (Board of Governors of the Federal Reserve System 2026b). To ensure that this package produces quality estimates, this report uses the package's functions to reproduce official FRB statistics. In general, the FRB uses a proprietary version of the data set to produce official statistics. The public-use data set is built to ensure respondent anonymity, and that set renders slightly different estimates from the proprietary set (Board of Governors of the Federal Reserve System 2026a). Fortunately, the FRB also publishes a version of its historical statistical tables using the public-use data set. If the procedures in the `scf` package are correct, then they should replicate the FRB's public data set-derived estimates. I find that the package's estimates are generally in agreement with the FRB's published estimates, with some exceptions that are generally very close (often to the second or third decimal place). The report offers a detailed comparison of the two sets of estimates. I maintain that the `scf` package produces correct univariate point estimates. I did not test the regression methods in this analysis.

These tables are:

- Table 1: Median and mean income by demographic characteristics
- Table 2: Pretax income by source, by net worth percentile
- Table 3: Reasons for saving by type of reason
- Table 4: Family net worth by selected demographic characteristics

I reproduce both the nominal (Board of Governors of the Federal Reserve System 2026d) and real dollar (Board of Governors of the Federal Reserve System 2026c) estimates, converting to nominal dollars with `scf_nominal()`. An earlier version of this validation exercise led to the development of `scf_pctile_sum()`, introduced in version 1.0.6 (Cohen 2026b). For further information on the package, see Cohen (2026c) and Cohen (2026a).

AI Use. The analysis used Claude in devising strategies for coding these analyses efficiently. The AI came up with good strategies quickly, which I vetted and tried to explain in the R Markdown file used to produce this report, linked above.

2 Methods

My strategy was to quantify the discrepancy between official statistics and those produced with the `scf` package. I quantify the percentage of cells within tolerance, and note the frequency and size of any discrepancies. I identify cells outside a tolerance of 0.1 percent for dollar figures and 0.05 points for percentages, which allows for rounding in the Fed's workbooks: dollar figures are rounded to the nearest $10 or $1, and percentages to two or three decimals. I also identify the largest discrepancies by table, statistic, and year.

The data are the files made by `scf_download()`, which hold the SCF's summary variables in 2025 dollars. The Fed's two workbooks are read from the folder named by the `docs_dir` parameter, or downloaded from the Federal Reserve if they are not there.

Each estimate comes from an `scf` function:

- Medians: `scf_median(method = "stack")`. The Fed stacks the five implicates and takes one weighted median. The default, `method = "implicate"`, averages the five implicate medians instead, which can differ in small groups (Cohen 2026b).
- Means: `scf_mean()`.
- Nominal dollars: `scf_nominal()`, applied to the median and mean results, which are in 2025 dollars.
- Percent of families in each group: `scf_freq()`.
- Percent that saved and reasons for saving: `scf_mean()` of the 0/1 indicators, times 100. The 1989 survey did not ask whether the family saved.
- Table 2's shares: `scf_ratio()` of each income source to the sum of the six sources. The Fed's `income` differs from that sum for many families, so dividing by `income` does not reproduce the table for the top net worth group.

Tables 1 and 4 report each statistic for all families and by nine characteristics, each a summary variable in the SCF: income percentile (`inccat`), age (`agecl`), family structure (`famstruct`), education (`edcl`), race or ethnicity (`racecl4`), work status (`occat1`), occupation (`occat2`), housing

status (`housecl`), and net worth percentile (`nwcat`). The income and net worth percentile groups are the Fed's own variables. In the Fed's sheets, the rows for each group sit at fixed places, in the order of the variable's codes, and the code reads them from those places. Each wave's estimates are saved, so later runs of the file reuse them.

3 Results

Table 1 summarizes agreement rates and largest gaps by the variable studied. Income and saving statistics, in the Fed's Tables 1 and 3, were above 99% in agreement; the income shares in the Fed's Table 2 were at 96%. Net worth estimates were 98% in agreement, but with considerable larger dollar discrepancies (maximum about 4%). The offending cells are listed below.

Table 1: Agreement by table and statistic

Table	Statistic	Workbook	Cells	Largest gap	Within
1	mean income	nominal	533	0.103%	99.8%
1	mean income	real	533	0.102%	99.8%
1	median income	nominal	533	0.007%	100.0%
1	median income	real	533	0.003%	100.0%
1	pct families	nominal	520	0.000 pt	100.0%
1	pct saved	nominal	492	0.018 pt	100.0%
2	pct of income	nominal	450	0.377 pt	96.2%
3	pct	nominal	117	0.000 pt	100.0%
4	mean net worth	nominal	533	1.047%	97.9%
4	mean net worth	real	533	3.313%	98.5%
4	median net worth	nominal	531	0.029%	100.0%
4	median net worth	real	531	3.829%	98.5%

By survey year (Table 2 below), there was 99% agreement in all years since 1995. In 1989 and 1992, around 3.5% of cells were discrepant.

Table 2: Agreement by survey year

Year	Cells	Largest dollar gap	Largest percent gap	Within
1989	412	0.875%	0.163 pt	96.6%
1992	453	0.514%	0.377 pt	96.5%
1995	454	1.799%	0.009 pt	99.3%
1998	454	0.229%	0.043 pt	99.6%
2001	453	3.313%	0.005 pt	99.6%
2004	450	0.230%	0.007 pt	99.6%
2007	454	0.186%	0.009 pt	99.8%
2010	450	0.039%	0.005 pt	100.0%
2013	450	0.042%	0.005 pt	100.0%
2016	452	3.829%	0.005 pt	99.8%
2019	452	0.838%	0.011 pt	99.8%
2022	453	0.104%	0.011 pt	99.6%
2025	452	0.110%	0.010 pt	99.6%

3.1 Cells outside tolerance

Most cells outside tolerance are medians and means of the bottom quarter of net worth. These are near zero, so the Fed's rounding to the nearest $10 is a large share of the value. Income shares by source in 1989 and 1992 are the other main group.

Table 3: Cells outside tolerance, Fed Table 1

Row	Year	Statistic	Workbook	Fed	scf	Gap
Other or Multiple Race	1989	mean income	nominal	33.338	33.304	-0.103%
Other or Multiple Race	1989	mean income	real	84.117	84.032	-0.102%

Table 4: Cells outside tolerance, Fed Table 2

Row	Year	Statistic	Workbook	Fed	scf	Gap
Business, NW group 5	1992	% of income	nominal	29.080	28.703	-0.377 pt
Wages, NW group 5	1992	% of income	nominal	43.274	43.509	0.235 pt
Transfers, NW group 3	1989	% of income	nominal	5.140	4.977	-0.163 pt
Transfers, NW group 4	1992	% of income	nominal	11.573	11.425	-0.148 pt
Wages, NW group 4	1992	% of income	nominal	63.913	64.025	0.112 pt
Wages, NW group 3	1989	% of income	nominal	76.100	76.205	0.105 pt
Transfers, NW group 2	1992	% of income	nominal	7.700	7.605	-0.095 pt
Transfers, NW group 1	1992	% of income	nominal	15.510	15.421	-0.089 pt
Transfers, NW group 3	1992	% of income	nominal	9.390	9.303	-0.087 pt
Capital gains, NW group 5	1989	% of income	nominal	13.109	13.189	0.080 pt
Wages, NW group 5	1989	% of income	nominal	43.283	43.204	-0.079 pt
Wages, all families	1992	% of income	nominal	62.942	63.017	0.075 pt
Wages, NW group 2	1992	% of income	nominal	81.130	81.205	0.075 pt
Wages, NW group 1	1992	% of income	nominal	74.290	74.362	0.072 pt
Business, all families	1992	% of income	nominal	13.232	13.172	-0.060 pt
Transfers, NW group 5	1992	% of income	nominal	5.880	5.934	0.054 pt
Wages, NW group 3	1992	% of income	nominal	72.550	72.601	0.051 pt

Table 5: Cells outside tolerance, Fed Table 4

Row	Year	Statistic	Workbook	Fed	scf	Gap
Net worth: less than 25	2016	median NW	real	0.130	0.135	3.829%
Net worth: less than 25	2001	mean NW	real	0.150	0.145	-3.313%
Net worth: less than 25	1995	mean NW	real	-0.310	-0.316	1.799%
Net worth: less than 25	2001	mean NW	nominal	0.080	0.079	-1.047%
Net worth: less than 25	1989	median NW	real	0.280	0.278	-0.875%
Net worth: less than 25	2019	median NW	real	0.390	0.393	0.838%
Net worth: less than 25	1995	mean NW	nominal	-0.150	-0.149	-0.672%
Net worth: less than 25	1992	mean NW	real	-1.270	-1.263	-0.514%
Net worth: less than 25	1992	mean NW	nominal	-0.560	-0.557	-0.499%
Net worth: less than 25	1989	mean NW	nominal	-0.540	-0.542	0.400%
Net worth: less than 25	1992	median NW	real	1.130	1.134	0.333%
Other or Multiple Race	1989	mean NW	nominal	164.050	164.457	0.248%
Other or Multiple Race	1989	mean NW	real	413.940	414.955	0.245%
Other not working	1989	median NW	real	1.460	1.463	0.236%
Net worth: less than 25	2004	mean NW	nominal	-1.330	-1.333	0.230%
Net worth: less than 25	1998	median NW	real	1.000	0.998	-0.229%
Net worth: less than 25	2007	mean NW	nominal	-2.110	-2.106	-0.186%
Net worth: less than 25	1989	mean NW	real	-1.370	-1.368	-0.148%
Net worth: less than 25	2004	median NW	real	2.920	2.923	0.112%
Net worth: less than 25	1998	mean NW	nominal	-1.850	-1.848	-0.112%
Net worth: less than 25	2025	mean NW	nominal	-3.930	-3.934	0.110%
Net worth: less than 25	2025	mean NW	real	-3.930	-3.934	0.110%
Net worth: less than 25	1995	median NW	real	2.010	2.012	0.109%
Net worth: 90–100	2022	mean NW	nominal	7771.160	7779.248	0.104%
Net worth: 90–100	2022	mean NW	real	8504.590	8513.441	0.104%
College degree	1989	mean NW	real	1035.440	1034.399	-0.101%
College degree	1989	mean NW	nominal	410.370	409.958	-0.100%

3.2 The 2025 survey

The tables below set the 2025 estimates beside the Fed's, in thousands of 2025 dollars. They show that measures of central tendency for income and net worth largely agree, and discrepancies are minor (generally to second decimal place).

Table 6: Family income, 2025

Group	Median, Fed	Median, scf	Mean, Fed	Mean, scf
All families	82.16	82.16	144.63	144.63
Income: less than 20	22.59	22.59	22.44	22.44
Income: 20–39.9	49.30	49.30	49.52	49.53
Income: 40–59.9	82.16	82.16	82.70	82.70
Income: 60–79.9	130.43	130.43	133.44	133.44
Income: 80–89.9	204.38	204.38	206.79	206.79
Income: 90–100	400.54	400.54	663.17	663.25
Less than 35	67.78	67.78	89.03	89.03
35–44	97.57	97.57	139.37	139.37
45–54	108.86	108.86	186.90	186.90
55–64	92.43	92.43	196.35	196.35
65–74	69.84	69.84	136.93	136.93
75 or more	66.76	66.76	124.79	124.79
Single with child(ren)	49.30	49.30	65.30	65.30
Single, no child, age less than 55	51.35	51.35	64.99	64.99
Single, no child, age 55 or more	45.19	45.19	71.79	71.79
Couple with child(ren)	128.38	128.38	210.69	210.69
Couple, no child	118.11	118.11	202.24	202.23
No high school diploma	35.95	35.95	52.26	52.26
High school diploma	60.59	60.59	82.72	82.72
Some college	69.84	69.84	95.85	95.85
College degree	129.41	129.41	231.44	231.43
White non-Hispanic	95.51	95.51	169.77	169.77
Black or African-American non-hispanic	49.30	49.30	73.55	73.55
Hispanic or Latino	66.76	66.76	94.73	94.72
Other or Multiple Race	81.14	81.14	148.34	148.33
Working for someone else	95.51	95.51	141.01	141.01
Self-employed	93.46	93.46	262.60	262.60
Retired	60.59	60.59	111.15	111.15
Other not working	43.13	43.14	86.00	86.01
Managerial or professional	120.16	120.16	220.64	220.63
Technical, sales, or services	77.03	77.03	113.07	113.08
Other occupation	81.14	81.14	99.12	99.13
Retired or other not working	56.49	56.49	107.35	107.35
Owner	108.86	108.86	183.65	183.65
Renter or other	50.32	50.32	70.33	70.33
Net worth: less than 25	39.03	39.03	49.18	49.18
Net worth: 25–49.9	67.78	67.78	79.72	79.71
Net worth: 50–74.9	99.62	99.62	109.59	109.59
Net worth: 75–89.9	152.00	152.00	184.26	184.26
Net worth: 90–100	309.13	309.14	573.41	573.48

Table 7: Family net worth, 2025

Group	Median, Fed	Median, scf	Mean, Fed	Mean, scf
All families	216.10	216.10	1232.00	1232.00
Income: less than 20	12.84	12.84	145.33	145.33
Income: 20–39.9	80.00	80.00	254.58	254.68
Income: 40–59.9	203.10	203.10	462.27	462.25
Income: 60–79.9	385.50	385.50	803.49	803.55
Income: 80–89.9	818.00	818.00	1421.45	1421.48
Income: 90–100	3624.60	3624.60	7565.79	7566.59
Less than 35	32.82	32.82	135.72	135.72
35–44	156.00	156.00	669.31	669.31
45–54	254.67	254.67	1243.61	1243.61
55–64	411.34	411.34	1871.02	1871.02
65–74	431.90	431.90	1870.92	1870.92
75 or more	504.60	504.60	1949.71	1949.71
Single with child(ren)	45.90	45.90	299.91	299.89
Single, no child, age less than 55	25.38	25.38	175.11	175.12
Single, no child, age 55 or more	192.30	192.30	830.44	830.49
Couple with child(ren)	333.60	333.60	1509.03	1509.01
Couple, no child	469.50	469.50	2152.94	2152.80
No high school diploma	18.70	18.70	293.88	293.98
High school diploma	106.71	106.71	470.03	470.11
Some college	181.50	181.50	661.35	661.35
College degree	505.03	505.03	2240.46	2240.36
White non-Hispanic	332.12	332.12	1682.38	1682.36
Black or African-American non-hispanic	36.42	36.42	242.49	242.49
Hispanic or Latino	79.90	79.90	334.84	334.85
Other or Multiple Race	205.00	205.00	1013.50	1013.47
Working for someone else	156.00	156.00	681.17	681.18
Self-employed	575.10	575.10	3469.98	3469.89
Retired	380.90	380.90	1485.52	1485.54
Other not working	16.90	16.90	437.60	437.63
Managerial or professional	295.30	295.30	1869.92	1869.85
Technical, sales, or services	151.57	151.57	567.08	567.11
Other occupation	108.16	108.16	453.55	453.58
Retired or other not working	297.60	297.60	1327.15	1327.19
Owner	459.00	459.00	1798.41	1798.41
Renter or other	10.55	10.55	153.53	153.53
Net worth: less than 25	1.69	1.69	-3.93	-3.93
Net worth: 25–49.9	100.50	100.50	109.09	109.08
Net worth: 50–74.9	405.82	405.82	432.37	432.37
Net worth: 75–89.9	1265.74	1265.74	1397.20	1397.25
Net worth: 90–100	4926.50	4926.50	8875.48	8875.93

4 Reproducing this report

Download the R Markdown file from https://github.com/jncohen/scf/tree/main/validation and install `scf` 1.2.0 or later. Download the data into a folder with `scf_download(seq(1989, 2025, 3))`, then knit the file with `rmarkdown::render("scf2025-validation.Rmd", params = list(data_dir = "path/to/data"))`, giving the folder that holds the data.

The first run estimates every wave and takes about an hour. Every cell, with the Fed's value, the `scf` estimate, and the difference, is written to `out/comparison.csv`.

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